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Subplate Neuron Dendritic Morphology During Development in Gyrencephalic and Lissencephalic Brains

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Abstract

Subplate neurons (SPNs) are among the earliest-born and most mature neurons in the developing cortex and are critical for thalamocortical axon pathfinding, radial migration, and early thalamocortical circuit formation. SPNs have been proposed to drive cortical folding, but their specific morphological features are not fully understood. This study examined SPN dendritic arborisation in two gyrencephalic species (sheep and ferret) before, during, and after the onset of primary folding, as well as at developmentally equivalent stages in the lissencephalic spiny mouse. Golgi-stained brains were analysed using Sholl analysis, along with additional metrics including number of primary branches, maximum intersections, path length, and skewness (branching asymmetry). Qualitative observations revealed region-specific differences in dendritic orientation and cell-type distribution between gyri and sulci in the gyrencephalic species, which were not found in the lissencephalic spiny mouse. MANOVA and two-way ANOVA with Bonferroni post-hoc tests showed no overall difference in SPNs arborisation throughout development in any species. Similarly, no differences were identified between gyral and sulcal SPNs in the ferret throughout gyrification, except for skewness in the sheep following the emergence of primary folds, where sulcal SPNs showed less branching away from the soma. These findings provide a valuable reference for future studies

investigating SPNs and early cortical circuit formation in both gyrencephalic and lissencephalic brains. Further studies using advanced imaging and analytical methods may uncover more subtle region-specific differences in dendritic architecture and neuronal orientation.

Introduction

Neurons in the cerebral cortex form distinct morphological characteristics during postnatal development via growth and pruning of axons, dendrites, and synapses. Understanding these morphological phenotypes is essential, as neurons play a key role in the structural and functional development of the brain¹. During early cortical development, the subplate (SP) layer and its resident subplate neurons (SPNs) play a pivotal role in orchestrating the events that shape the formation of the cerebral cortex². The SP layer contains an extracellular matrix enriched with axon guidance and growth molecules, including draxin, proteoglycans, and

Unc5C, creating a microenvironment that supports the early ingrowth of thalamocortical axons³. Resident SPNs form transient synapse-like contacts with these afferents and display early electrical activity mediated by connexin hemichannels, enabling calcium-dependent communication with surrounding glia, such as astrocytes^{4,5}. In addition, migrating neurons briefly interact with SPNs along their route, receiving signals that regulate migration and early differentiation⁶. Together, these features position SPNs as active organizers within a transient glial–extracellular matrix niche, coordinating early neuronal migration, synaptogenesis, and the initial assembly of cortical circuits^{7,8}.

Although SPNs are essential for cortical formation, the cellular mechanisms underlying gyrification remains largely unknown, and their specific role in this process is yet to be determined. The SP layer, located beneath the developing cortical plate (CP), has been proposed to act as a driver of gyrification, the process that gives rise to a folded cortex in gyrencephalic species³. SPNs are among the earliest-maturing neurons and are fundamental for several key neurodevelopmental processes, including axonal pathfinding, neuronal migration, and the establishment of laminar structure⁷, all of which are also critical to healthy gyrification⁹.

In contrast to the lissencephalic (smooth) brain, gyrencephalic cortices are subjected to mechanical forces associated with folding that influence: (i) the absolute and relative thickness of cortical layers in the folds of the brain, called gyri and sulci, (ii) the trajectory and accumulation of migrating neurons, leading to a higher neuronal density in gyral peaks, and (iii) region-specific neuronal morphology. For example, gyral pyramidal neurons tend to be more elongated and follow the curvature of the fold, whereas those beneath sulci are more compressed, with shorter axons and distinct dendritic architecture¹⁰. This mechanical

environment, characterised by stretching in gyri and compression in sulci, may contribute to the emergence of distinct functional properties in these regions, as supported by functional magnetic resonance imaging studies showing divergent activity patterns between gyral and sulcal regions in the mature cortex¹¹. It has been proposed that early morphological differences in dendritic structure between gyral and sulcal neurons may underpin these functional specialisations by establishing region-specific neuronal properties^{11,12}.

Dendritic morphology is known to shape neuronal function and network dynamics. For example, recent studies have shown that changes in dendritic arbor size can induce a switch in neuronal network behaviour from synchronised to desynchronised activity, suggesting that even subtle alterations in dendritic structure may lead to significant differences in network-level dynamics^{13,14}. Moreover, slight changes in dendritic morphology such as changes in length, branching or orientation, can profoundly affect the attenuation of excitatory postsynaptic potentials, ultimately resulting in the generation of different types of action potentials^{10,15}.

SPNs are known to promote synchronous neuronal firing, which facilitates early network communication and supports the formation of cortical circuits¹⁶. The dendrites of SPNs extend into developing cortical layers during gyrification and act as a relay for both thalamocortical and corticocortical signals^{17,18}. Therefore, it is plausible that the SP, and the resident SPNs, may play a role in the formation of gyri and sulci^{3,19}. Notably, the SP layer is markedly thinner in lissencephalic species than in gyrencephalic species³. In species with a folded brain, the SP layer tends to be thicker and more structurally complex, potentially reflecting adaptations to support increased connectivity demands and dynamic activity patterns associated with

gyrification^{3,19}. This raises the possibility that the dendritic morphology of SPNs may be influenced by, or contribute to, the mechanical and cellular dynamics of cortical folding.

If regional differences in SPN morphology exist between gyri and sulci, they may represent early specific morphological specialisations that contribute to the functional divergence observed between these regions in the mature brain. SPNs have been described previously in the fetal cortex of human and monkey both during gyrification²⁰⁻²² and after the majority of gyrification has occurred²³⁻²⁵. However, the dendritic developmental trajectory of SPNs in relation to gyral and sulcal development remains largely unexplored. Consequently, how SPN morphology evolves in relation to cortical folding remains unanswered.

This study aimed to determine whether SPNs located beneath gyri and sulci exhibit uniform or region-specific dendritic morphology during cortical folding. In addition, we sought to characterise how SPN dendritic morphology changes over time, independent of regional location. Based on prior findings that neuronal morphology in gyral regions of the adult macaque is more elaborate than in sulcal regions¹⁰, we hypothesised that SPNs beneath gyri would exhibit more complex arborisation than those beneath sulci as gyrification progresses. To test this, we analysed SPN morphology in two gyrencephalic species (sheep and ferret) across key stages of cortical folding. In contrast, we hypothesised that SPN morphology would remain uniform throughout the cortex in the lissencephalic spiny mouse.

Results

Qualitative assessment of SPNs in gyrencephalic and lissencephalic species

Qualitative analysis of SPN morphology was conducted by using a combination of complexin 3 (CPLX3) and nuclear receptor related 1 (Nurr1) immunostaining with Golgi staining. In the

SP of pre-gyrencephalic sheep (gestational age [GA] 70; Fig. 1a, b) and ferret (postnatal day [P] 5; Fig. 2a, b), SPNs appeared immature, often lacking distinct dendritic or axonal structures. These cells exhibited a lamellipodia-like morphology with no clear preference for an established neuronal subtype (Supplementary Table 1; Supplementary Fig. 1, 7).

As gyrification progressed, SPNs in both sheep (GA80 – GA110; Fig. 1c-h) and ferret (P15 – P25; Fig. 2c-f) developed more defined dendritic and axonal processes. SPNs at this stage were classified as pyramidal, inverted and horizontal pyramidal, bipolar, multipolar and polymorphic, with the exception of GA80, where most SPNs still appeared immature, and very few defined neuronal subtypes could be identified (Supplementary Table 1; Supplementary Fig. 2).

In gyral SP at GA90 and GA110 (sheep), and P15 and P25 (ferret), SPNs frequently exhibited radial orientation toward the pial surface. At this age, SPN morphology was pyramidal, the most prominent subtype, with some inverted pyramidal, bipolar, and multipolar neurons. Horizontal pyramidal and polymorphic neurons appeared in comparatively smaller proportions (Supplementary Table 2; Supplementary Fig. 3, 5, 8, 10). In contrast, sulcal SPNs were predominantly tangentially oriented to the pial surface, with horizontal pyramidal morphology the most prevalent. In smaller proportions, we also observed a variety of bipolar, multipolar, and polymorphic neurons, with pyramidal and inverted pyramidal neurons appearing in the lowest proportions (Supplementary Table 2; Supplementary Fig. 4, 6, 9, 11, 13).

In the lissencephalic spiny mouse, SPN morphology remained consistent across developmental stages (Fig. 3a-c). Multipolar neurons represented the highest proportion of SPN subtypes at all developmental stages examined, followed by mixture of pyramidal, horizontal, inverted,

bipolar, and polymorphic neurons, but without any apparent orientation-specific pattern. This is unlike gyrencephalic species, in which the orientation of SPNs is strongly aligned within gyral (radial) and sulcal (horizontal) regions. These findings support the notion that region-specific variation in SPN morphology is a distinctive feature of gyrencephalic brain development. By contrast, our results demonstrate that SPNs of the lissencephalic spiny mouse display no clear orientation preference (Supplementary Table 3; Supplementary Fig. 12).

SPNs do not differ in dendritic complexity between gyrencephalic and lissencephalic brains

Repeated-measures analysis of variance (ANOVA) revealed a significant main effect of distance from the soma on the number of dendritic intersections across all species and developmental ages, reflecting the expected decrease in branching with increasing distance from the soma and the resultant positive skew in the data. A significant interaction between convolution type (gyrus vs sulcus) and distance from the soma was found only in sheep at GA90 ($F(29, 232) = 1.56, p = .04$; Supplementary Table 4). However, post-hoc analysis did not detect significant differences in the number of intersections between gyri and sulci at any specific distance (Fig. 1i-l).

In sheep, no significant interaction between distance from the soma and convolution, nor any significant main effect of convolution, was observed in the number of dendritic intersections at GA70 (pre-gyri/pre-sulci), GA80, or GA110 (Fig. 1i-l; Supplementary Table 4). These findings indicate that dendritic complexity of SPNs does not significantly differ beneath gyri and sulci during these stages of cortical development. Similarly, no significant differences in the number of intersections were found in the ferret (Fig. 2g-i; Supplementary Table 5) or spiny mouse (Fig. 3d-f; Supplementary Table 6) at any developmental age analysed.

Some morphological features of SPNs differ between gyri and sulci in gyrencephalic species

In the sheep, two-way multivariate ANOVAs (MANOVA) revealed no significant interaction effect between region (gyrus vs sulcus) and age for additional Sholl metrics (number of primary branches, maximum number of intersections, path length, and skewness; $F(12, 87) = 0.90, p = .55$; Supplementary Table 7). However, significant main effects of age were observed for all metrics, whereas no main effect of region was detected (Fig. 4).

The number of primary branches significantly changed as cortical folding progressed (Fig. 4a). Post-hoc analyses showed that beneath gyri, primary branch number was significantly lower at GA70 when compared to all ages, and at GA80 was significantly higher when compared to GA90 and GA110, while no differences were found between GA90 and GA110. In contrast, beneath sulci, the number of primary branches at GA70 was significantly lower than all other ages. The number of primary branches was greater at GA80 when compared to GA90 and GA110, and greater at GA90 compared to GA110, however, no significant differences were observed. For both gyri and sulci, the maximum number of dendritic intersections was significantly lower at GA70 compared to all other ages (Fig. 4b); but no significant differences were found between GA80, GA90, and GA110. Similarly, path length did not differ significantly between GA70 and GA80 but increased significantly from GA70 to GA90 and GA110 for both gyri and sulci. Path length also increased significantly from GA80 to GA90 and from GA80 to GA110, but not from GA90 to GA110 (Fig. 4c; Supplementary Table 10).

Finally, gyral SPN skewness was significantly lower at GA70 and GA80 compared to GA90 and GA110, indicating increased positive asymmetrical branching (inferred by number of

intersections across radii away from the soma) with age (Supplementary Table 8). In contrast, sulcal SPN skewness increased significantly only between GA70 and GA110, with no significant differences among GA80, GA90, and GA110 (Fig. 4d). Although no significant main effect of region was observed in the additional Sholl metrics, skewness trended towards significance ($F(1, 30) = 3.36, p = .07$; Supplementary Table 7). Exploratory pairwise comparisons revealed a single significant difference at GA110 ($p = .02$; Supplementary Table 9), with gyral SPNs exhibiting greater positive asymmetry (greater number of branches away from the soma) than sulcal SPNs (fewer number of branches away from the soma) at the end of gyrification.

In ferrets, no significant interaction between region and age was detected for any Sholl metric ($F(8, 40) = 0.66, p = .72$; Supplementary Table 7), although significant main effects of age were observed for the number of primary branches, maximum number of intersections, and path length, but not for skewness (Fig. 5; Supplementary Table 7). No significant regional differences were found between gyri and sulci (Supplementary Table 10). Beneath gyri, the number of primary branches significantly decreased from P5 to P15, but not between P5 and P25, or P15 and P25 (Fig. 5a). The maximum number of intersections differed only between P15 and P25 (Fig. 5b), while path length increased significantly from P5 to P25 and from P15 to P25 (Fig. 5c). Neither the number of primary branches nor skewness differed significantly across ages (Fig. 5a, d). In contrast, beneath sulci, both the maximum number of intersections and path length increased significantly from P5 to P25 and from P15 to P25 (Fig. 5b-c; Supplementary Table 11).

In the spiny mouse, no significant differences in the additional Sholl metrics were detected across regions of interest (ROIs) at E35 ($F(12, 45) = 1.22, p = .30$), P1 ($F(12, 45) = .70, p =$

.75), or P5 ($F(12, 45) = 0.98, p = .49$; Supplementary Table 12). To further examine potential developmental trends, average metric values were plotted to visualise mediolateral patterns across ages (Fig. 6). A marked increase in the number of primary branches, maximum number of intersections, and path length was observed from E35 to P1, followed by a modest decline from P1 to P5. In contrast, skewness values approached zero at P5 relative to E35 and P1, indicative that there was no preference of dendritic branching towards or away from the soma at this later developmental stage.

Dendritic spine density and type are comparable between mature gyral and sulcal SPNs

Linear spine density and spine type were assessed in SPNs from sheep at GA110 and ferrets at P25 to characterise dendritic spine morphology at the end stage of gyrification. The spiny mouse was not included in this analysis, as this investigation was preliminary and focused on comparing SPNs located beneath gyri and sulci in gyrencephalic species.

No significant differences in linear spine density were found between gyri and sulci in either sheep (Fig. 7a) or ferret (Fig. 7d). Similarly, analysis of spine subtype distribution revealed no significant differences in linear density across spine types between gyri and sulci in both species (Fig. 7b-c; e-f; Supplementary Table 13). In both sheep and ferret, thin spines were the most prevalent subtype on SPNs, while bifurcated spines were the least frequently observed (Supplementary Table 14).

Discussion

This study aimed to determine whether SPNs exhibit conserved or divergent dendritic morphologies across species with different patterns of gyrification, and to characterise their development during key stages of neurodevelopment in both gyrencephalic and lissencephalic

brains. Our qualitative observations revealed distinct regional SPNs morphologies in gyral and sulcal areas of gyrencephalic species, a pattern not observed in the lissencephalic spiny mouse, where SPNs exhibited mature-like features from early developmental stages. In contrast to these qualitative observations, quantitative analyses revealed no significant difference in dendritic branching complexity between gyral and sulcal SPNs in either gyrencephalic species or across equivalent dorsolateral regions in the spiny mouse. However, significant age-related differences were observed. In sheep, sulcal SPNs exhibited decreased skewness with age, whereas gyral SPN exhibited increased skewness and with more branching away from the soma. This divergence in branching was not observed in ferrets, where SPNs in both regions displayed similar asymmetric branching by the end of gyrification. These findings suggest that dendritic branching may be region-specific and species-dependant, with sheep sulcal SPNs exhibiting different branching skewness patterns during folding. Spine analyses at the end of gyrification (GA110 in sheep, P25 in ferret) revealed no regional differences in total spine density or distribution of spine types. Thin and filopodia spines, indicative of immature neurons, were the most prevalent in both species. These dynamic spine types are associated with early synaptogenesis and structural plasticity, suggesting that SPNs remain in an immature phase of spinogenesis at the completion of primary cortical folding.

Prior to the onset of gyrification (GA70 in sheep; P5 in ferret), SPNs predominantly displayed immature, polymorphic morphologies with lamellipodia-like processes and are sparsely distributed, consistent with descriptions of early migratory neurons^{26,27}. Similarly, a lower density of neurons has been reported in the ferret cortex at P7 underneath layer VI (presumably the SP)²⁸. Previously, Rapid Golgi staining performed in the pre-gyrencephalic human visual cortex between 12-24GW detected neurons confined to the marginal zone and CP²⁹. In that study, the SP contained only sparse immature neurons²⁹, consistent with our observations in

pre-gyrencephalic sheep and ferret (GA70/P5). Conversely, SPNs in the spiny mouse displayed clearer and earlier maturation, with well-defined dendrites and axons already evident at E35, an age approximating the early gyrification period in ferrets and sheep (Supplementary Fig. 14). By P1 and P5, spiny mouse SPNs exhibited similar morphologies. Notably, Golgi-impregnated SPNs at E35 (term = 38–40 days)³⁰ appeared more mature than those described in the E17.5 conventional laboratory mouse (term $\approx$ 20 days) using fixed-tissue Golgi staining³¹.

Indeed, some studies have reported mature-like SPNs during early pre-gyrencephalic stages. For example, Rapid Golgi performed on formalin-fixed GA71 sheep cortical tissue revealed bipolar and pyramidal neurons within the SP³². Similar observations have been reported in human and monkey cortex using both unfixed and fixed tissue with the Stensaas-modified Del Río-Hortega method, as well as fixed tissue with Rapid Golgi²⁰⁻²². Additionally, other studies have employed variations of Golgi protocols, including Rapid Golgi and Del Río-Hortega-based techniques on fixed fetal monkey tissue^{33,34}, as well as modified Rapid Golgi protocols in the cat^{35,36}, and have likewise reported well-differentiated neurons in the early SP, both prior to and at the onset of gyrification.

These contrasting observations may reflect differences in Golgi impregnation technique, fixation protocol, developmental timing, or a combination of these factors. Early embryonic tissue is particularly prone to poor impregnation with the Rapid Golgi method when processed fresh³⁵, which can lead to an under-representation of neurons in transient layers such as the SP. The Rapid Golgi method used in the present study is a modified Golgi–Cox technique that preferentially impregnates more mature neurons³⁷. However, this limitation is partially mitigated when the protocol is applied to formalin-fixed tissue, which enhances staining

reliability in early developmental stages^{37,38}. In contrast, other Golgi staining approaches, such as the Stensaas-derived method, may be more effective for preferentially labelling deeper and more mature neurons within the SP at these early stages.

Comparisons with other unfixed Rapid Golgi studies further support our interpretations. In mice, neurons in the cerebral cortex visualised with Rapid Golgi were largely reported later in development at P7³⁹, and in the rat entorhinal cortex at P7, where layer VI neurons still appeared poorly differentiated⁴⁰. Given that Rapid Golgi preferentially impregnates more mature neurons, the earlier appearance of well-defined SPNs in the spiny mouse suggests that these neurons develop and mature embryonically to a greater extent than in conventional rodent species.

Additionally, an earlier study using the Stensaas-modified Del Río-Hortega method in rats identified a range of SPN morphologies at E18, including pyramidal, horizontal, and polymorphic neurons, extending through P1⁴¹. We now consider the possibility that alternative Golgi-staining methods with greater sensitivity to immature neurons could be advantageous in the spiny mouse and may enable SPN maturation to be traced even earlier in gestation. Our observations suggest that SPNs in the spiny mouse likely undergo substantial embryonic differentiation and maturation compared with typical rodent models. Applying other staining methods in future work would be particularly valuable for precisely tracking the onset and progression of SPN maturation.

Although the neurodevelopmental timeline of the spiny mouse is not yet fully characterised, our data provide new insights into early SPN development and maturation in this species. SPNs in the spiny mouse brain exhibited more mature morphologies, including multipolar, horizontal

bipolar, and pyramidal forms, much earlier than typically reported in the developing human or mouse cortex^{42,43}. Importantly, our findings highlight the utility of Golgi staining for identifying neuronal maturity and provide an important resource for future studies targeting key neurodevelopmental milestones.

Early maturation of SPN morphology was observed in sheep from GA80. Mature morphological features included a defined soma with emerging dendritic or axonal processes, although most neurons still appeared relatively immature, with numerous short neurites. In contrast, by P15 in the ferret, SPNs displayed markedly more mature morphologies, characterised by well-defined dendrites and axons. This difference likely reflects species-specific variation in developmental timing. In sheep, primary gyrification progresses over an extended gestational interval, beginning around GA70 and continuing until approximately GA110, spanning nearly 40 days^{3,32,44-46}. In ferrets, primary folds emerge postnatally, around P5, and are largely established by P28, a period of roughly 23–25 days⁴⁷⁻⁴⁹. Importantly, when comparing equivalent temporal windows, approximately ten days after the onset of gyrification (GA80 in sheep and P15 in ferrets), SPNs in the ferret appeared substantially more mature than those in sheep, as revealed by Rapid Golgi staining. This suggests that SPNs in ferrets may undergo more rapid dendritic and axonal differentiation during the early phase of gyrification. As a result, reaching comparable neurodevelopmental milestones takes longer in sheep. The slower maturation of SPNs in large-brained mammals may reflect a structural constraint, as the ratio of cortical thickness to SP thickness is greater, which has been proposed to influence both the pace of SP development and the timing of neuronal maturation³.

During the mid-to-late stages of gyrification in the ferret (P15 and P25) and sheep (GA90 and GA110), gyral SPNs predominantly exhibited pyramidal, inverted, multipolar, and bipolar

morphologies, with horizontal and polymorphic forms being least common. In contrast, sulcal SPNs were mainly horizontal, bipolar, multipolar, pyramidal, and polymorphic, while horizontal-pyramidal neurons were observed least frequently. Our data demonstrates that distinct SPN subtypes are differentially distributed beneath gyri and sulci, and that this is likely a feature of gyrencephalic brain development. In contrast, regional morphological distinctions were not observed during comparable neurodevelopmental ages of lissencephalic development.

Previously, several studies have qualitatively reported that SPNs (and layer VIb neurons) beneath gyri tend to be “normal” and inverted pyramidal, multipolar or bipolar, whereas those located at the base of sulci are predominantly horizontal and bipolar^{42,50,51}. Consistent with these earlier reports, gyral SPNs from ferret and sheep were predominantly pyramidal, whereas sulcal SPNs displayed horizontal pyramidal morphology, confirming a regional preference in neuronal type.

Distinct differences in SPN morphology and cell type distribution between gyral and sulcal regions may support their role in cortical fold formation. Given that SPNs relay electrical activity to the CP and organise cortical circuitry during development¹⁶, our observations support two possibilities: SPNs either actively contribute to the formation of gyri and sulci, or their orientation and morphology are shaped by their final position within a gyrus or sulcus. The first scenario suggests that maturational and morphological differences in SPNs could prime and macroscopically shape the developing cortex, potentially predicting gyral and sulcal formation before it occurs. However, the second scenario appears more likely, as our qualitative observations indicate that SPNs are morphologically similar between regions in the pre-gyrified brain. Divergence in SPN morphology and maturation emerges during mid-to-late folding, suggesting that SPNs may not directly initiate cortical folds. This implies that

differences in SPN morphology and orientation may arise in response to the mechanical forces generated during gyrification. Nevertheless, this does not exclude the possibility that SPNs contribute functionally throughout gyrification, and early differences may have gone undetected due to the limitations of the Rapid Golgi technique.

We performed quantitative Sholl analysis to assess potential morphological differences between these regions. Unexpectedly, the number of dendritic intersections as a function of distance from the soma did not differ significantly between gyral and sulcal SPNs across species or developmental stages. Of the morphological parameters assessed, only skewness exhibited a significant association with gyrification, as demonstrated by significant main effects of age in both gyral and sulcal regions.

In sheep, overall skewness increased in both gyral and sulcal SPNs with development, but not differing by region until GA110. At GA110, sulcal SPNs exhibited lower skewness value of 0.4, indicating that their dendritic branching remained more concentrated toward the soma, compared to gyral SPNs which exhibited higher skewness value of 0.8. The data suggests that gyral SPNs extend a greater proportion of their dendritic branching further from the soma, whereas sulcal SPNs retain more of their branching closer to the soma. Skewness difference in the sheep at GA110 could be due to the neuronal subtypes present between gyral (pyramidal) and sulcal (horizontal pyramidal) regions. However, a similar subtype distribution was also present in the ferret at P15 and P25, yet no skewness difference was observed. Given the timing-related developmental differences previously observed between sheep and ferrets, it is possible that such regional differences in dendritic branching may emerge later in ferret cortical development.

When comparing additional morphological metrics across development in gyral and sulcal regions, both shared and region-specific patterns emerged. In sheep, the number of SPN primary branches (neurites extending directly from the soma) increased from GA70 to GA80, despite cells retaining an immature-like morphology. This increase may reflect increased migratory activity during this developmental window, consistent with reports that immature, migratory neurons exhibit a dynamic neuritic behaviour, frequently extending and retracting their processes⁵². Thus, the early increase in primary branching may represent an adaptive response to facilitate the influx of migratory neurons into the developing cerebral cortex.

At mid-gyrification (GA90 in sheep, P15 in ferret), the number of primary branches decreased, coinciding with SPN maturation. This transition from an immature to a mature state is characterised by the emergence of a distinct soma, dendrites, and a clearly defined axon. Observations across both species suggest a conserved trajectory of primary branch formation during gyrification. However, no significant differences were observed between gyral and sulcal regions, indicating that primary branching alone is unlikely to drive regional morphological specialisations during cortical folding.

The maximum number of intersections, representing peak dendritic branching, generally followed the trajectory of primary branch development. During early stages (GA70-80 in sheep, P5 in ferret), values reflected initial neurite outgrowth, while later stages (GA90-110 in sheep, P15-25 in ferret), showed increased complexity. In sheep, sulcal SPNs showed a progressive increase in peak intersections, whereas gyral SPNs remained relatively stable. In contrast, both gyral and sulcal SPNs in ferrets showed a significant increase in the number of intersections by late gyrification. These findings suggest more extensive dendritic elaboration in sulcal SPNs, though differences between regions did not reach statistical significance.

Path length increased significantly in SPNs from GA80 to GA90 and again to GA110 in sheep, and from P15 to P25 in ferrets, with no differences between gyral and sulcal regions. In sheep, this increase in path length coincided with the transition from immature to a mature neuronal morphology between GA80 and GA90. In contrast, ferret SPNs appeared morphologically mature by P15 but did not exhibit a significant increase in path length until P25. This suggests species-specific timing of dendritic growth relative to gyrification: in sheep, maturation and dendritic elongation occurred concurrently around 20 days after the onset of gyrification (~GA90), whereas in ferrets, maturation occurred earlier (P15, ~10 days after the onset of gyrification), with dendritic elongation following even later at P25 (~20 days post-onset). Given that Rapid Golgi preferentially stains mature neurons³⁷, these findings may highlight divergent trajectories of dendritic maturation and growth, with ferret SPNs maturing prior to dendritic elongation, while sheep SPNs undergo both processes in parallel.

The inclusion of additional metrics complements the primary Sholl analysis, which alone cannot capture all aspects of dendritic morphology. For example, path length measurements provide information on neuronal size, an element not addressed by Sholl analysis. These supplementary metrics revealed a comparable pattern of dendritic growth in neuronal size and preferential branching patterns across cortical regions in two gyrencephalic species, despite differences in developmental trajectories, such as growth rates, and brain size⁵³⁻⁵⁵.

Taken together, these findings suggest that (i) throughout gyrification, SPNs exhibit a distinct orientation and cell preference according to their position relative to gyri or sulci, while exhibiting similar dendritic branching except for differences in branching patterns; (ii) in sheep, sulcal SPNs exhibit dendritic branching skewness closer to the soma when compared to gyral

SPNs, a pattern not observed in ferrets (with less complex cortical folding) or the smooth-brained spiny mouse; and (iii) during maturation of the gyrified brain, sulcal and gyral SPNs share developmental trajectories linked to age, characterised by comparable primary branching, peak dendritic intersections, and overall neuronal size.

In our analysis of spine type and density, the predominant spine types observed in GA110 sheep and P25 ferrets were thin and filopodia spines, which are considered immature⁵⁶ and highly dynamic, frequently extending and retracting to form connections with presynaptic terminals. This indicates that spinogenesis is in its early stages during primary fold formation, at least in SPNs. Early in brain development, neurons typically exhibit filopodia and thin spines, which contribute significantly to rapid structural plasticity and the formation of new synaptic connections⁵⁷. Spine morphology generally shifts from filopodia and thin spines to mushroom spines as the brain matures, reflecting a transition from less stable spines (filopodia without synapses and thin spines with unstable synapses) to more stable mushroom spines^{27,56}.

Our study additionally characterised spine features in SPNs at the end of gyrification (i.e., the end of primary folding) and found that spine number and type did not differ between regions (gyrus vs sulcus). One limitation of this analysis was that, due to the low number and limited diversity of spines at earlier ages in ferret and sheep samples, faithful quantification could not be performed. This may reflect a limitation in the sensitivity of the Rapid Golgi method rather than a true biological absence, indicating that alternative techniques may be required to examine early spines. Indeed, other studies have reported the presence of spines at the onset of gyrification in fetal sheep and humans using modified Golgi methods^{23,34}. Applying such alternative staining approaches at these early stages would provide important insight into the

trajectory of SPN spinogenesis throughout gyrification, including the emergence of filopodia-like protrusions and their transition to more mature spine forms.

Our findings do not support the initial hypothesis that gyral SPNs would exhibit greater dendritic complexity during gyrification, as previously observed in the CP at GA90 in sheep⁴⁵ and in the adult monkey brain¹⁰. Instead, our findings demonstrate that, throughout gyrification, overall dendritic branching was comparable between gyral and sulcal SPNs, despite qualitative differences in orientation and regional cell preference. In contrast, the spiny mouse showed uniformity in dendritic branching, orientation, and cell preference across the cortical mantle, while displaying mature-like features consistent with later developmental stages observed in rats and mice^{31,39,40}. Across developmental stages, both gyral and sulcal SPNs followed a similar trajectory in primary branching, peak dendritic complexity, and overall path length. However, sulcal SPNs showed distinct differences in dendritic branching away from the soma relative to gyral SPNs, suggesting region-specific developmental differences not evident in the lissencephalic spiny mouse. Furthermore, at the end of gyrification, spine density in both sheep and ferret SPNs were characteristic of immature neuronal types, with no significant regional differences observed between gyral and sulcal regions.

Despite the known drawbacks and caveats of Golgi staining of vast neuronal subtypes, it remains an effective tool for visualising detailed dendritic architecture in a subset of neurons without significant overlap, allowing quantitative assessment of large neuronal populations. While a substantial number of neurons were analysed, subtle morphological differences may require a larger sample size for detection. Alternatively, the absence of observed differences may reflect limitations of the Sholl method itself. Although widely used, Sholl analysis measures dendritic complexity based on the number of intersections between dendrites and

concentric circles radiating from the soma, potentially missing branches that fall between radii and underestimating dendritic complexity. Additionally, it does not account for other important aspects of neuronal morphology, such as overall neuron size, the number and length of primary branches, angular distribution, orientation relative to the pial surface, or dendritic directionality.

The absence of detectable dendritic differences does not exclude the possibility of functional contribution of SPNs to gyrification, given the accumulating evidence implicating the SP in early circuit formation, axonal organisation, and increasingly, cortical folding itself^{3,44,58,59}. Importantly, determining whether gyral and sulcal SPNs differ in their connectivity or causally influence cortical development in region-specific ways will require approaches beyond morphology alone. Intracellular dye injections or similar methods to reconstruct full dendritic arbors and trace their projections to specific cortical layers, together with functional analyses to assess whether gyral and sulcal SPNs integrate or transmit activity in a unique fashion, will be essential to resolve this question. Thus, the potential region-specific involvement of SPNs in gyrification remains an important area for future investigation.

Collectively, our findings provide a comparative framework and valuable resource for future investigations of SPNs and early circuit formation in both gyrencephalic and lissencephalic brains. The application of techniques capable of resolving finer aspects of dendritic architecture and orientation may further uncover region-specific features that distinguish gyral and sulcal SPNs.

Methods

Animal ethics and housing

All procedures involving animals were approved by the relevant institutional ethics committees: the Monash University Animal Ethics Committee (application MMCA/2019/10 for sheep), and the RMIT University Animal Ethics Committee (applications AEC1933 for spiny mice and AEC1906 for ferrets).

Developmental Timepoints

Animals were collected at developmental stages corresponding to key phases of cortical maturation. Timepoints that represent three major neurodevelopmental milestones in each species: the onset of neurogenesis, the period of cortical folding (in gyrencephalic species), and the initiation of myelination is found in Supplementary Fig. 14.

For the gyrencephalic species, tissue was obtained prior to the onset of gyrification (Sheep: GA70; Ferret: P5), during gyrification (Sheep: GA80 and GA90; Ferret: P15), and at the completion of primary fold formation (Sheep: GA110; Ferret: P25)^{3,32,44-49,53-55}. For the lissencephalic spiny mouse, equivalent neurodevelopmental stages were assessed (E35, P1, and P5)^{30,60-64}.

Tissue Collection

Date-mated pregnant Merino x Border Leicester ewes ($n = 20$) carrying twin fetuses were humanely euthanised via intravenous administration of sodium pentobarbitone (130 mg/kg; Lethobarb, Virbac Pty. Ltd., Australia), and fetuses ($n = 40$) were delivered via Caesarean section. Ferret kits at P5, P15, and P25 ($n = 10$ per age) were euthanised using deep anaesthesia with isoflurane inhalation followed by decapitation. For spiny mice at E35, pregnant dams ($n = 5$) were euthanised with an overdose of sodium pentobarbitone (200-250 mg/kg; i.p), and fetuses ($n = 10$) were delivered via Caesarean section. Neonatal spiny mice (P1 and P5; $n = 10$

per age) were euthanised with sodium pentobarbitone (200 mg/kg; i.p) followed by decapitation.

At each developmental stage, the brain from one fetal sheep twin (GA70, GA80, GA90, GA110, $n = 5$ per age), ferret kits (P5, P15, P25, $n = 5$ per age), and fetal and neonatal spiny mice (E35, P1, P5, $n = 5$ per age) were extracted from the skull and hemisected (spiny mouse), or hemisected and divided into 5mm-thick coronal blocks or regions (sheep and ferret) and processed for Golgi staining.

For histological and immunohistochemical analysis, brains from the other fetal sheep twin (GA70, GA80, GA90, GA110, $n = 5$ per age) and ferret kits (P5, P15, P25, $n = 5$ per age) were transcardially perfused with 0.9% saline (pH 7.4) followed by 4% paraformaldehyde (PFA; 0.1 M, pH 7.4), and post-fixed in 4% PFA for 24 to 48 hours. Spiny mice brains (E5, P1, P5, $n = 5$ per age) were immersion-fixed in 4% PFA (pH 7.4) for 24 to 48 hours. Following fixation, brains were hemisected (spiny mouse), or hemisected and divided into 5mm-thick coronal blocks or regions (sheep and ferret; Supplementary Fig. 15), placed in tissue processing cassettes (Tissue-Tek, USA), and processed into paraffin wax using an automatic tissue processor (Leica ASP200S).

Golgi staining

One case (GA70 sheep) was excluded because tissue quality after processing was insufficient for reliable Golgi staining and reconstruction. Sex distribution of animals and animal number used for Golgi staining in this study can be found in Supplementary Table 15.

Golgi staining was performed using the FD Rapid GolgiStain Kit™ (FD Neurotechnologies, USA) according to the manufacturer's instructions on fresh tissue blocks selected from region 2 (R2) in the sheep, ferret, and spiny mouse brains (see Supplementary Fig. 15). Brain tissue was incubated in impregnation solution AB for 14 days, with a solution change after the first 24 hours. Following impregnation, solution AB was replaced with solution C (clearing solution) for 7 days, again with a change of solution after the first 24 hours. Cleared tissue was frozen using double-distilled water on dry ice and sectioned coronally at 100µm thickness using a cryostat (-24°C to -26°C). Sections were mounted on 3% gelatine-coated slides with a drop of solution C and air-dried in the dark at room temperature for three days. Slides were washed twice in milli-Q water, stained with a mixture of solution D and E, and washed twice in distilled water. Sections were dehydrated through graded ethanols (50%, 70%, and 100%), cleared in histolene, and coverslipped using DPX mounting medium (Merck, Australia). Stained sections were stored at room temperature, protected from light, until visualisation.

Histology and Immunohistochemistry

Paraffin-embedded blocks from sheep, ferret, and spiny mouse brains were coronally sectioned at 8µm thickness using a rotary microtome (Jung Biocut 2035, Geprüfte Sicherheit, Germany). From the R2 block, two consecutive serial sections were collected at every 200µm interval, mounted on charged glass slides (Superfrost Plus, Menzel Glaser, Germany), and air-dried at room temperature for 24-48 hours.

To confirm the location of the SP in Golgi-stained sections of each species, corresponding cortical region sections were immunostained using SP-specific markers Nurr1 and CPLX3. Antigen retrieval was performed in citrate buffer (pH 6.0) via microwave heating. Endogenous peroxidase activity was quenched using 3% H₂O₂ in Tris-buffered saline (TBS), and

nonspecific binding was blocked with DAKO serum. Sections were incubated overnight at 4°C with either rabbit anti-Nurr1 (1:1000, Santa Cruz Biotechnology) or rabbit anti-CPLX3 (1:4000, Synaptic Systems), followed by biotinylated goat anti-rabbit IgG secondary antibody (1:200, Vector Laboratories) and the avidin-biotin complex (ABC Elite kit, Vector Laboratories). Immunoreactivity was visualised using 3,3'-diaminobenzidine (DAB; MP Biomedicals). Sections were counterstained, dehydrated through a graded ethanol series, cleared in histolene, and coverslipped using DPX. Omission of the primary antibody served as a negative control. For Nissl Staining, Sections were dewaxed in two changes of histolene (10-15 dips), rehydrated in graded ethanol solutions (2 x 100%, 2 x 70%, 2 x 50% ethanol; 10-15 dips each), washed in dH₂O for one minute, and stained with 0.1% cresyl violet in dH₂O (C5042, Sigma Aldrich, MO, USA) for 1-2 minutes. Sections were washed once in dH₂O (one minute), dehydrated in graded ethanol solutions (2 x 50%, 2 x 70%, 2 x 100% ethanol; 10-15 dips each), coverslipped using DPX mounting medium (Merck, Germany).

Region of Interest (ROI) Selection

In the GA70 sheep and P5 ferret brains, where gyri and sulci were not yet clearly defined, ROIs were selected based on early morphological landmarks. 'Pre-sulci' were defined as developing indentations (ROI2 and ROI4), while 'pre-gyri' referred to elevated regions likely to develop into gyri or located adjacent to sulci (ROI1, ROI3 and ROI5) (Supplementary Fig. 16). ROIs were placed as evenly as possible along the cortical mantle (Supplementary Fig. 16a, d). During and after the onset of gyrification (GA80, GA90, and GA110 for sheep; P15 and P25 for ferret), gyri and sulci were clearly identifiable, allowing for accurate placement of ROIs (labelled as 'G' for gyri and 'S' for sulci; Supplementary Fig. 16b, c, e). In the lissencephalic spiny mouse, four equidistant ROIs (ROI1 to ROI4) were placed mediolaterally along the cortical mantle at all developmental stages (Supplementary Fig. 16f).

Morphological assessment of SPNs

The positioning and laminar distribution of SPNs was assessed at all developmental stages in fetal sheep, ferret, and spiny mouse brains using Nurr1 or CPLX3 immunostaining. Immunolabelled sections were scanned using the Aperio ScanScope AT Turbo eSlide scanner, and high-power images were obtained with Aperio ImageScope freeware (Leica Biosystems). Qualitative and quantitative analyses of SPN morphology across all ages and species were conducted on Golgi-stained sections imaged using a Nikon Confocal C1 microscope at 20x magnification. Image acquisition and analysis were performed using NIS-Elements software (v4.30.02, Build 1053, Nikon Instruments Inc.).

SPNs were classified based on qualitative morphological features, using criteria adapted from previous studies on layer VIb neurons^{14,20-25,32,42,65-67}. The number of neurons included in this qualitative analysis are outlined in Supplementary Tables 1-3. Neurons were visually inspected and grouped into neuronal classes based on defining morphological characteristics of a particular class.

The neuronal classes were defined as follows:

1. Pyramidal neuron: Characterised by a triangular-shaped soma, a single apical dendrite extending toward the cortical surface, and multiple basal dendrites.
2. Horizontal pyramidal neuron: A subtype of pyramidal neuron orientated parallel to the cortical layers, with dendrites and axon projecting laterally rather than vertically toward the cortical surface.
3. Inverted pyramidal neuron: A pyramidal neuron with an inverted orientation, such that the apical dendrite extends toward the deeper layers instead of the cortical surface.

4. Multipolar neuron: Exhibits multiple dendritic processes extending from the soma in various directions, lacking a distinct apical dendrite.
5. Bipolar neuron: Displays an elongated, spindle-shaped soma and two primary dendrites extending from opposite ends.
6. Neurogliaform neuron: Resembles glial cells in appearance, with a rounded soma and numerous short, straight processes extending from the cell body. This neuron class was not included in the qualitative analysis.
7. Polymorphic neuron: Defined as mature cells that do not fit any morphological category and displays irregular soma shape with dendrites emerging in multiple directions.

Sholl analysis of SPNs

For Sholl analysis, z-stacks of Golgi-stained sections were captured, comprising 15-25 z-steps (or optical sections) at 1 μ m intervals to encompass the entire dendritic arbor of each SPN. Neurons were included for Sholl analysis based on the following criteria: (i) sufficient Golgi stain impregnation, (ii) intact dendritic architecture, and (iii) no overlap between dendrites of adjacent neurons. The preferred sampling location was the upper SP–CP border, which can be reliably identified using Nurr1/CPLX3. Moreover, at developmental stages where Golgi-stained neurons were sparse (GA70 in sheep and P5 in ferret), and SPNs were absent at the upper CP–SP border, neurons were selected from the lower SP–IZ border. The CP–SP and SP–IZ boundaries were verified using matched Nissl- and Nurr1/CPLX3- immunostained sections from equivalent ages and regions. This ensured accurate layer identification and minimized the inclusion of migrating cells from adjacent compartments. In all cases, neurons were selected from within the SP, with preference given to the most mature cells, consistent with literature indicating that at these early developmental stages SPNs represent the earliest-maturing neuronal population^{16,21}. Concentric circles were automatically generated at 10 μ m intervals

from the centre of the soma, and the number of dendritic intersections at each interval was recorded. For each species, developmental age, and ROI (i.e., pre-sulcus, pre-gyrus, sulcus, and gyrus in sheep and ferret, and ROIs 1-4 in spiny mice), the mean number of intersections at each radial distance from the soma was calculated. A total of 670 SPNs from sheep, 541 from ferret, and 300 from spiny mouse were included in this analysis.

Additional Metrics of Dendritic Complexity

In addition to the standard Sholl analysis, complementary metrics were extracted using the Simple Neurite Tracer plugin in ImageJ. These metrics are derived from the Sholl data and provide further insight into dendritic arborisation patterns at varying distances from the soma. Unlike the main Sholl profiles, which pool intersection counts across all neurons within each gyral or sulcal group at every radius step, these additional metrics were calculated from each neuron individually and then averaged per animal. This approach produced a single value per animal for each measure, allowing a complementary assessment of dendritic complexity independent of radius-based pooling. For all additional metrics, final sample sizes for the number of neurons included were as follows: sheep = 570 SPNs, ferret = 430 SPNs, and spiny mouse = 300 SPNs. A description of the additional metrics is provided below.

Number of primary branches: Primary branches refer to dendrites that originate directly from the soma and serve as a fundamental measure of neuronal complexity and potential synaptic integration. This metric can also be used to determine neuronal types⁶⁸. However, Sholl analysis does not inherently differentiate between primary, secondary, and tertiary branches⁶⁹. Unless each branch is explicitly traced and classified using the Simple Neurite Tracer plugin, a labour-intensive process when applied across large datasets, intersections at each radius are aggregated irrespective of branch origin. This limitation may obscure differences in dendritic

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structure. For example, two neurons may show similar numbers of intersections at a given radius yet differ in branching pattern: one neuron might have fewer but more extensive primary branches, while another has many shorter, more distal branches.

Maximum number of intersections: This metric represents the peak number of dendritic intersections at a single radial distance from the soma and serves as an index of maximal dendritic complexity. It enables comparison between neurons based on the location of their most complex arborisation point, independent of distance from the soma.

Path length: Path length refers to the cumulative length of all neurite branches traced from a single neuron. Because Sholl analysis involves manual skeletonisation, ImageJ automatically provides the total path length of the traced structure. This parameter has been used in previous studies to describe “total neuronal wiring” in three-dimensional reconstructions⁷⁰ and is a useful proxy for overall dendritic elaboration.

Skewness: Skewness provides a quantitative measure of how data points are distributed around their mean. In the context of Sholl analysis, this reflects the distribution of the number of intersections or branching across radial distances, providing insight into how dendritic complexity is organised along the proximal-distal axis of the neuron. Skewness indicates whether values are extended toward larger or smaller magnitudes relative to the mean. As described in the ImageJ Sholl plugin documentation, positive skewness (right-skewed distribution) indicates a longer tail extending toward higher values, whereas negative skewness (left-skewed distribution) indicates a longer tail extending towards lower values. As a general guideline, skewness values greater than +1.0 or less than -1.0 are typically considered clearly asymmetrical. Thus, in Sholl analysis, skewness quantifies both the direction and extent of the

tail in the distribution of intersections across radial distances from the soma. Here, the 0 radius corresponds to the soma, the number of intersections reflects dendritic branching, and increasing radii indicate branching occurring progressively farther from the soma.

A more positively skewed (right-skewed) distribution reflects a stronger distal bias in dendritic branching, with intersections extending farther from the soma. Conversely, a less positively skewed distribution indicates a weaker distal bias, with branching occurring at comparatively shorter distances from the soma. This metric has previously been used to characterise morphological differences in microglia under both physiological and pathological conditions⁷¹.

Dendritic spine density

Dendritic spines were counted along a 50µm segment of the apical dendrite of SPNs located in both gyral and sulcal regions. One neuron per sulcus and one per gyrus was analysed from a single section per animal, using matched Golgi-stained sections from GA110 sheep ($n = 5$) and P25 ferrets ($n = 5$). Imaging was performed using an Olympus BX61 microscope at 100x magnification with oil immersion and a z-step interval of 0.2µm. For each animal, five sulcal SPNs and five gyral SPNs were analysed. The mean number of spines per 10µm dendritic segment was calculated for each animal, and group means were obtained by averaging the individual animal means⁷².

SPNs were selected for spine analysis according to the following criteria: (i) a clearly identifiable soma with an apical dendrite oriented perpendicular to the pial surface; (ii) complete Golgi impregnation sufficient to visualise of the full extent of the dendritic arbor, and (iii) unobstructed dendritic processes, free from overlap with neighbouring neurons, glial cells, or blood vessels. Basal dendrites were excluded from analysis due to the large degree of overlap

with other dendrites, which hindered accurate spine quantification. In addition, spine data were stratified by morphological spine subtype (e.g., mushroom, thin, stubby, filopodia, and bifurcated) according to established classifications⁷³.

Statistical analysis

Normality of all datasets was assessed using the Shapiro-Wilk normality test. Sholl intersection data violated assumptions of normality across all groups ($p < .05$). Two-way repeated measures ANOVA was used for these data as there is no suitable non-parametric alternative, and ANOVAs are robust to violations of normality⁷⁴.

Normality was confirmed for all additional Sholl analyses ($p > .05$), with the exception of the following groups: maximum number of intersections in GA70 sheep sulci, GA80 sheep gyri, and skewness in P15 ferret sulci ($p < .05$). Spine density data in both ferret and sheep satisfied the assumption of normality, however, spine type data were not normally distributed and were therefore analysed using the non-parametric Wilcoxon Signed-Rank test. All statistical analyses were performed using GraphPad Prism (v9.0.1, San Diego, CA, USA) or IBM SPSS Statistics (v27, Armonk, NY, USA). Statistical significance was defined as $\alpha < .05$.

To assess differences in Sholl metrics across all species and developmental stages, two-way repeated-measures ANOVAs were conducted, with multiple pairwise comparisons with Bonferroni adjustment. For gyrencephalic species (sheep and ferret), the within-subject factor was distance from the soma (i.e., number of dendritic intersections per radius), and the between-subject factor was convolution (gyri vs sulci; Supplementary Tables 4 and 5). For the lissencephalic spiny mouse, distance from the soma was treated as the within-subject factor, and cortical region (ROIs 1-4) as the between-subject factor (Supplementary Table 6).

To further assess group difference in dendritic morphology, two-way MANOVAs were performed to compare additional Sholl metrics (i.e., number of primary branches, maximum number of intersections, path length, and skewness) between gyri and sulci in sheep and ferret (Supplementary Table 7). Age was not included as a factor in spiny mouse analyses, as cortical regions (ROIs 1–4) could not be accurately matched across developmental stages in the same manner as gyri and sulci in gyrencephalic species. Instead, separate one-way MANOVAs were performed for each developmental stage in spiny mice (Supplementary Table 12). To explore, mediolateral differences across development, average values for each ROI were plotted and compared. Where significant simple main effects were observed, Bonferroni-adjusted post hoc pairwise comparisons were conducted (Supplementary Tables 8–11).

To compare linear spine density between gyri and sulci in sheep and ferret, paired t-tests were performed. Differences in spine subtype distributions (mushroom, thin, stubby, filopodia, and bifurcated) were assessed using Wilcoxon Signed-Rank tests due to non-normal distributions. Descriptive statistics, and statistical outcomes for spine density and spine subtype analyses are reported in Supplementary Tables 13 and 14.

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Statement of Ethics. All tissue collection was performed in accordance with procedures approved by the RMIT University Animal Ethics Committee (Approvals AEC1933; AEC1906)

and the Monash University Animal Ethics Committee (Approval: MMCA/2019/10). This study was performed in accordance with ARRIVE guidelines.

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Data Availability Statement. The data that support findings of this study are included in this article and its supplementary material files. Further enquiries can be directed to the corresponding author.

Figure Legends

Figure 1. Immunostaining, Golgi-staining, and Sholl analysis of gyral and sulcal SP in the sheep brain across gyrification. (a–h) Immunostaining of the SP marker CPLX3, except at GA70 in (a) pre-gyral and (b) pre-sulcal regions, where Nurr1 was used, alongside corresponding Golgi-stained photomicrographs: (a–b) GA70 (before gyrification), (c–d) GA80 (mid-gyrification), (e–f) GA90 (mid-gyrification), and (g–h) GA110 (late-gyrification). (i–l) Sholl analysis at (i) GA70, (j) GA80, (k) GA90, (l) GA110 comparing gyral and sulcal SPNs. The y-axis represents the number of dendritic intersections; the x-axis indicates distance from the soma. Pre-gyral/gyral SPNs are shown as black circles; pre-sulcal/sulcal SPNs as pink squares. Data were analysed using two-way repeated-measures ANOVA with Bonferroni post hoc correction. Values are presented as mean $\pm$ 95% CI. Each data point represents the mean number of intersections at each radius, pooled across all animals within the gyral or sulcal group, with 15–20 neurons analysed per animal. At GA70 (a–b), SPNs in pre-gyral and pre-sulcal regions appear less differentiated and lack clear orientation, though they remain consistently located beneath the CP. In contrast, SPNs at GA80 (c–d), GA90 (e–f), and GA110 (g–h), display more mature morphologies with voluminous cytoplasm and distinct orientation patterns. In gyral regions (c, e, g), SPNs exhibit radial or slightly tangential orientation, whereas in sulcal regions (d, f, h) orientation is predominantly horizontal or slightly tangential. Dotted squares (a–h) indicate magnified insets; dotted red square indicates the region corresponding to the Golgi-stained photomicrographs; dotted lines delineate the SP–CP boundary. Maximum distances from the soma were defined by the longest neuron analysed: GA70 = 120 μ m, GA80 = 110 μ m, GA90 = 300 μ m, GA110 = 410 μ m. For shorter neurons, intersections beyond their length were recorded as zero. Abbreviations: CP, cortical plate; GA, gestational age; SP, subplate; SPNs, subplate neurons. Scale bars = 200 μ m (immunostained)/100 μ m (Golgi-stained).

Figure 2. Immunostaining, Golgi-staining, and Sholl analysis of gyral and sulcal SP in the ferret brain across gyrification. (a–f) Immunostaining of the SP marker CPLX3 in: (a) pre-gyral and (b) pre-sulcal regions at P5 (prior to gyrification), with corresponding Golgi-stained photomicrographs; (c–d) P15 (mid-gyrification); (e–f) P25 (late-gyrification) in gyral (c, e) and sulcal (d, f) regions. (g–i) Sholl analysis comparing dendritic intersections of gyral (black circles) and sulcal (pink squares) SPNs at (g) P5, (h) P15, and (i) P25. The y-axis represents the number of dendritic intersections; the x-axis indicates distance from the soma. Data were analysed using two-way repeated-measures ANOVA with Bonferroni post hoc corrections and are presented as mean $\pm$ 95% CI. Each data point represents the mean number of intersections at each radius, pooled across all animals within the gyral or sulcal group, with 15–20 neurons analysed per animal. At P5 (a–b), SPNs in pre-gyral and pre-sulcal regions appear less differentiated and show no clear orientation, though they are consistently located below the CP. By P15 (c–d) and P25 (e–f), SPNs exhibit more mature morphologies with voluminous cytoplasm. In gyral SP (c, e) CPLX3⁺ and Golgi-stained SPNs show radial or slightly tangential orientation, while sulcal SP (d, f), orientation is predominantly horizontal or slightly tangential. Dotted squares in (a–f) indicate locations of magnified insets; dotted red square indicates the region corresponding to the Golgi-stained photomicrographs; dotted lines delineate the SP–CP boundary. Maximum distances from the soma were determined by the longest neuron analysed at each age: P5 = 230 μ m, P15 = 200 μ m, P25 = 240 μ m. Intersections beyond these maxima were recorded as zero for shorter neurons. Abbreviations: CP, cortical plate; P, postnatal; SP, subplate; SPNs, subplate neurons. Scale bars = 200 μ m (immunostained)/100 μ m (Golgi-stained).

Figure 3. Immunostaining, Golgi-staining, and Sholl analysis of SPNs in the lissencephalic spiny mouse. (a–c) Immunostaining of the SP marker CPLX3 in the (a) E35, (b) P1, and (c)

P5 spiny mouse brain, with corresponding Golgi-stained photomicrographs. (d–f) Sholl analysis of SPNs at (d) E35, (e) P1, and (f) P5 across the cortical mantle: ROI1 (black circles), ROI2 (pink squares), ROI3 (green triangles), and ROI4 (inverted purple triangles). The y-axis indicates the number of dendritic intersections; the x-axis shows distance from the soma across four mediolateral regions (ROI1 to ROI4). Data were analysed using two-way repeated-measures ANOVA with Bonferroni post hoc correction and are presented as mean $\pm$ 95% CI. Each data point represents the mean number of intersections at each radius, pooled across all animals within the gyral or sulcal group, with 20 neurons analysed per animal. In immunostained sections, the SP appears as a thin band located beneath the subcortical white matter and above the CP at all developmental stages. Golgi-stained SPNs exhibited a range of morphologies, including bipolar, pyramidal, inverted pyramidal, and multipolar neurons at all time points. Dotted squares in (a–c) indicate locations of magnified insets; dotted line delineates the approximate extent of the SP. Maximum distance from the soma was determined by the longest neuron analysed at each age: E35 = 100 μ m, P1 = 170 μ m, P5 = 170 μ m. For neurons shorter than these maxima, intersections beyond their length were recorded as zero. Abbreviations: CP, cortical plate; E, embryonic; P, postnatal; SP, subplate; SPNs, subplate neurons; SCWM, subcortical white matter. Scale bars = 200 μ m (immunostained)/100 μ m (Golgi-stained).

Figure 4. Quantitative analysis of additional Sholl metrics for gyral and sulcal SPNs in the developing fetal sheep brain at GA70, GA80, GA90, and GA110. (a) Number of primary dendritic branches, (b) maximum number of dendritic intersections, (c) path length, and (d) skewness of intersection distribution in gyral (black circles) and sulcal (red squares) SPNs. Each data point represents the mean of 15 neurons per group per animal. Data are presented as mean $\pm$ SD. * p < .05, ** p < .01. Black arrow ($\rightarrow$) indicates a significant effect across

developmental ages; ‘#’ denotes a significant difference between gyral and sulcal regions.

Abbreviations: GA, gestational age; SPNs, subplate neurons.

Figure 5. Quantitative analysis of additional Sholl metrics for gyral and sulcal SPNs in the developing ferret brain at P5, P15, and P25. (a) Number of primary dendritic branches, (b) maximum number of dendritic intersections, (c) path length, and (d) skewness of intersection distribution in gyral (black circles) and sulcal (red squares) SPNs. Each data point represents the mean of 15 neurons per group per animal. Data expressed as mean $\pm$ SD. * $p < .05$, ** $p < .01$. Black arrow ($\leftarrow$) indicates a significant effect across developmental ages. Abbreviations: P, postnatal; SPNs, subplate neurons.

Figure 6. Quantitative analysis of additional Sholl metrics for SPNs in the spiny mouse at E35, P1, and P5. (a) Number of primary dendritic branches, (b) maximum number of dendritic intersections, (c) path length, and (d) skewness of intersection distribution in SPNs from ROI1 (black circles), ROI2 (red squares), ROI3 (green inverted triangles), and ROI4 (purple triangles). Each data point represents the mean of 15 neurons per group per animal. Data are presented as mean $\pm$ SD. Abbreviations: E, embryonic; P, postnatal; ROI, region of interest; SPNs, subplate neurons.

Figure 7. Linear dendritic spine density and spine type analysis of gyral and sulcal SPNs in the sheep and ferret at the end of gyrification. (a) Average linear spine density (spines/50 μ m) in gyral and sulcal SPNs in GA110 sheep, expressed as mean $\pm$ SD. (b) Spine type distribution (linear density) in gyral and sulcal SPNs in GA110 sheep, expressed as median $\pm$ 95% CI. (c) Bright-field photomicrographs of Golgi-stained gyral and sulcal SPNs in GA110 sheep showing dendritic spines (100x magnification; colour inverted in ImageJ). (d) Average

linear density in gyral and sulcal SPNs in P25 ferret, expressed as mean $\pm$ SD. (e) Spine type distribution in gyral and sulcal SPNs in P25 ferret, expressed as median $\pm$ 95% CI. (f) Bright-field photomicrographs of Golgi-stained gyral and sulcal SPNs in P25 ferret (100x magnification; colour inverted in ImageJ). Abbreviations: GA, gestational age; P, postnatal; SPNs, subplate neurons. Scale bars = 10 μ m.

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